

Strategic Investment with Asymmetric Information, Jeffrey Updating, and Endogenous Volatility

A Self-Contained Model for Symbolic Solution

Contents

1	Overview and Motivation	2
2	Primitives and Notation	2
2.1	Players and Budgets	2
2.2	Return Processes — Binomial Mechanism	2
2.3	Public History	3
2.4	Fee Structure	3
2.5	Allocation Decision	3
3	Belief Structure — Beta-Binomial Model	3
3.1	Prior	3
3.2	Bayesian Posterior from History	4
3.3	Marketing and the Jeffrey Shift	4
3.4	Perceived Drift and Volatility	5
4	Terminal Wealth and Preferences	5
4.1	Terminal Wealth	5
4.2	Mean-Variance Objective	6
4.3	Computing Mean and Variance	6

5	Stackelberg Equilibrium	6
5.1	Useful Shorthand	7
5.2	Step 1 — Follower’s Problem (Player 2)	7
5.2.1	Optimal allocation α_2^*	7
5.2.2	Optimal marketing m_2^*	7
5.3	Step 2 — Leader’s Problem (Player 1)	8
5.3.1	Optimal allocation α_1^*	8
5.3.2	Derivative of follower’s allocation with respect to m_1	8
5.3.3	First-order condition for m_1^*	9
5.4	Equilibrium Summary	10
6	Realised Portfolio Volatility	10
6.1	Equilibrium Variance Expression	10
6.2	Comparative Statics on Realised Volatility	10
6.2.1	Effect of marketing m_i^*	10
6.2.2	Effect of α_j^*	11
6.2.3	Effect of fee f	11
7	Structural GARCH Representation	11
7.1	Multi-Period Extension	11
7.2	Period t Surprise	11
7.3	Belief Dynamics	11
7.4	Propagation to Realised Variance	12
7.5	GARCH(1,1) Representation	12
7.6	Endogenous GJR Asymmetry	13
7.7	Stationarity Condition	13
8	Complete Parameter List	14
9	Guide to Symbolic Solution in SymPy / Mathematica	16
9.1	Step-by-Step Recipe	16
9.2	Key Equation Cross-Reference	17
10	Comparative Statics Summary	18

1 Overview and Motivation

This document presents a complete, self-contained model of strategic investment between two asymmetrically informed players. Each player owns a risky stochastic process and may invest in the other's process for a fee. Players market themselves to attract investment, and beliefs about the other's process are updated via Jeffrey conditioning applied to a Beta-Binomial model of returns.

The key outputs of the model are:

- (i) Equilibrium allocations α_i^* and marketing spends m_i^* in closed form.
- (ii) An endogenous expression for perceived return volatility as a function of model primitives.
- (iii) A structural GARCH(1,1) representation of realised portfolio volatility, with endogenously derived GJR-type asymmetry.

All objects are defined sufficiently explicitly to permit full symbolic solution in SymPy or Mathematica.

2 Primitives and Notation

2.1 Players and Budgets

There are two players, indexed $i \in \{1, 2\}$, with j denoting the opponent of i , i.e. $j = 3 - i$.

Definition 2.1 (Budgets). *Player i begins with initial wealth $W_i > 0$. Players may have asymmetric budgets: $W_1 \neq W_2$ in general. Player i chooses a marketing spend $m_i \geq 0$ with $m_i \leq W_i$, leaving an effective investment budget of*

$$W_i^{\text{eff}} = W_i - m_i.$$

2.2 Return Processes — Binomial Mechanism

Definition 2.2 (Binomial Return Process). *Each period t , player i 's process generates a return*

$$r_i^t = \begin{cases} u_i & \text{with (true) probability } p_i, \\ d_i & \text{with (true) probability } 1 - p_i, \end{cases}$$

where $u_i > d_i$ are fixed, publicly known constants (the up and down returns), and $p_i \in (0, 1)$ is the true up-probability, known only to player i .

Draws are i.i.d. across periods. The true mean and variance of r_i^t are:

$$\mu_i = p_i u_i + (1 - p_i) d_i, \quad (1)$$

$$\sigma_i^2 = p_i(1 - p_i)(u_i - d_i)^2. \quad (2)$$

Remark 2.1. *Both μ_i and σ_i^2 are determined entirely by the single parameter p_i given fixed u_i, d_i . This is the key parsimony of the binomial mechanism.*

2.3 Public History

Definition 2.3 (Observed History). *Both players observe the same public history of n past discrete periodic returns from each process. Let k_i denote the number of up-moves observed in player i 's history:*

$$\mathbf{r}_i = (r_i^1, \dots, r_i^n), \quad k_i = \#\{t : r_i^t = u_i\}.$$

The sample frequency is $\bar{p}_i = k_i/n$. The history is objective and agreed upon by both players.

2.4 Fee Structure

Definition 2.4 (Performance Fee). *When player i invests in player j 's process, they receive the return net of a proportional performance fee $f \in (0, 1)$. Player j retains fraction f of the gross return as fee income. The fee is fixed and exogenous.*

2.5 Allocation Decision

Definition 2.5 (Allocation). *Player i chooses $\alpha_i \in [0, 1]$, the fraction of W_i^{eff} allocated to player j 's process. The remaining fraction $(1 - \alpha_i)$ stays in player i 's own process. Incoming investment from player j is accepted passively — there is no cost to receiving investment.*

3 Belief Structure — Beta-Binomial Model

3.1 Prior

Player j holds a prior belief about player i 's up-probability p_i . Since $p_i \in [0, 1]$ and the return mechanism is Binomial, the natural conjugate prior is:

Definition 3.1 (Beta Prior).

$$p_i \sim \text{Beta}(\alpha_i^0, \beta_i^0),$$

where $\alpha_i^0 > 0$ and $\beta_i^0 > 0$ are prior pseudo-counts representing player j 's initial beliefs. The prior mean is $\alpha_i^0/(\alpha_i^0 + \beta_i^0)$.

3.2 Bayesian Posterior from History

Proposition 3.1 (Bayesian Update). *After observing k_i up-moves in n periods, the Bayesian posterior is:*

$$p_i \mid \mathbf{r}_i \sim \text{Beta}(\alpha_i^*, \beta_i^*),$$

where

$$\alpha_i^* = \alpha_i^0 + k_i, \quad \beta_i^* = \beta_i^0 + n - k_i.$$

The posterior mean and variance are:

$$\mathbb{E}[p_i \mid \mathbf{r}_i] = \frac{\alpha_i^*}{\alpha_i^* + \beta_i^*}, \quad (3)$$

$$\text{Var}[p_i \mid \mathbf{r}_i] = \frac{\alpha_i^* \beta_i^*}{(\alpha_i^* + \beta_i^*)^2 (\alpha_i^* + \beta_i^* + 1)}. \quad (4)$$

Since history is public, both players agree on this posterior.

3.3 Marketing and the Jeffrey Shift

Player i spends $m_i \geq 0$ on marketing to signal that their drift μ_i is higher than the history alone suggests. This constitutes *soft evidence* in the sense of Jeffrey (1965): it shifts player j 's credence over p_i without being a clean Bayesian observation.

Definition 3.2 (Marketing Effectiveness Function). *The marketing spend m_i generates a pseudo-count shift:*

$$\delta(m_i) = \frac{m_i}{m_i + \gamma_i}, \quad \gamma_i > 0,$$

where γ_i is a scale parameter governing the rate of diminishing returns. Note $\delta(0) = 0$, $\delta \in (0, 1)$, $\delta'(m_i) > 0$, $\delta''(m_i) < 0$, and $\lim_{m_i \rightarrow \infty} \delta(m_i) = 1$.

Definition 3.3 (Jeffrey-Shifted Posterior). *The Jeffrey update on the Beta posterior adds $\delta(m_i)$ pseudo up-counts:*

$$\alpha_i^{**} = \alpha_i^* + \delta(m_i), \quad \beta_i^{**} = \beta_i^*.$$

Player j 's operational (post-Jeffrey) belief about p_i is:

$$p_i \sim \text{Beta}(\alpha_i^{**}, \beta_i^{**}).$$

Remark 3.1 (Rigidity Condition). *Jeffrey's rule requires that the conditional distribution of returns given p_i is unaffected by the soft evidence (marketing). This holds here: $\Pr(r_i^t = u_i \mid p_i) = p_i$ is determined by the process, not by marketing. Marketing only shifts credence over which p_i is the true one.*

3.4 Perceived Drift and Volatility

Definition 3.4 (Perceived Drift and Volatility). *Player j 's Jeffrey-updated belief about p_i implies the following perceived drift and volatility for player i 's process. Define:*

$$\hat{p}_i^j \equiv \mathbb{E}^j[p_i] = \frac{\alpha_i^{**}}{\alpha_i^{**} + \beta_i^{**}} = \frac{\alpha_i^* + \delta(m_i)}{\alpha_i^* + \delta(m_i) + \beta_i^*}.$$

Then:

$$\mu_i^j = \hat{p}_i^j \cdot u_i + (1 - \hat{p}_i^j) \cdot d_i, \quad (5)$$

$$\sigma_i^{j2} = (u_i - d_i)^2 \cdot \hat{p}_i^j (1 - \hat{p}_i^j). \quad (6)$$

Remark 3.2. *Both μ_i^j and σ_i^{j2} are functions of m_i through $\delta(m_i)$. Marketing simultaneously shifts the perceived drift and perceived volatility. The volatility is maximised at $\hat{p}_i^j = 1/2$ and decreasing on both sides.*

The full expression for perceived volatility in terms of model primitives is:

$$\boxed{\sigma_i^{j2} = (u_i - d_i)^2 \cdot \frac{(\alpha_i^* + \delta(m_i)) \beta_i^*}{(\alpha_i^* + \delta(m_i) + \beta_i^*)^2}} \quad (7)$$

4 Terminal Wealth and Preferences

4.1 Terminal Wealth

Definition 4.1 (Terminal Wealth). *Player i 's terminal wealth at horizon T is:*

$$W_i(T) = R_i \cdot A_i + R_j \cdot B_i, \quad (8)$$

where R_i is the gross realised return from player i 's process (mean μ_i , variance σ_i^2), and:

$$A_i = (1 - \alpha_i)W_i^{\text{eff}} + f \alpha_j W_j^{\text{eff}}, \quad (9)$$

$$B_i = \alpha_i W_i^{\text{eff}} (1 - f). \quad (10)$$

Here A_i is player i 's effective exposure to their own process (own investment plus fee income from player j 's investment), and B_i is their net exposure to player j 's process after the fee.

4.2 Mean-Variance Objective

Definition 4.2 (Mean-Variance Utility). *Each player maximises a mean-variance objective:*

$$\mathcal{V}_i = \mathbb{E}[W_i(T)] - \frac{\rho_i}{2} \text{Var}[W_i(T)], \quad (11)$$

where $\rho_i > 0$ is player i 's coefficient of absolute risk aversion.

4.3 Computing Mean and Variance

Since R_i and R_j are independent:

$$\mathbb{E}[W_i(T)] = \mu_i A_i + \mu_j^i B_i, \quad (12)$$

$$\text{Var}[W_i(T)] = A_i^2 \sigma_i^2 + B_i^2 \sigma_j^{i2}, \quad (13)$$

where μ_j^i and σ_j^{i2} are player i 's Jeffrey-updated beliefs about player j 's process (symmetric to equations (5)–(6) with indices swapped).

Note that player i knows their own true μ_i and σ_i^2 exactly, but uses Jeffrey-shifted beliefs μ_j^i , σ_j^{i2} for player j 's process.

Expanding the full objective:

$$\begin{aligned} \mathcal{V}_i = & \mu_i [(1 - \alpha_i)W_i^{\text{eff}} + f \alpha_j W_j^{\text{eff}}] + \mu_j^i \alpha_i W_i^{\text{eff}} (1 - f) \\ & - \frac{\rho_i}{2} \left[((1 - \alpha_i)W_i^{\text{eff}} + f \alpha_j W_j^{\text{eff}})^2 \sigma_i^2 + (\alpha_i W_i^{\text{eff}} (1 - f))^2 \sigma_j^{i2} \right]. \end{aligned} \quad (14)$$

5 Stackelberg Equilibrium

The two players play a Stackelberg game. Player 1 is the *leader* (justified by $W_1 > W_2$ without loss of generality) and moves first, choosing (m_1, α_1) . Player 2 observes

m_1 , applies the Jeffrey update, and then chooses (m_2, α_2) . The equilibrium is solved by backward induction.

5.1 Useful Shorthand

For each player i with opponent j , define:

$$\Delta_i = \mu_j^i(1-f) - \mu_i \quad (\text{net return differential}), \quad (15)$$

$$\Sigma_i = \sigma_i^2 + (1-f)^2 \sigma_j^{i2} \quad (\text{total perceived risk}), \quad (16)$$

$$F_i = W_i^{\text{eff}} + f \alpha_j W_j^{\text{eff}} \quad (\text{fee-augmented budget}). \quad (17)$$

5.2 Step 1 — Follower's Problem (Player 2)

Player 2 takes (m_1, α_1) as given and solves:

$$\max_{\alpha_2, m_2} \mathcal{V}_2.$$

5.2.1 Optimal allocation α_2^*

Taking the first-order condition of $\mathcal{V}_2$ with respect to α_2 :

$$\frac{\partial \mathcal{V}_2}{\partial \alpha_2} = W_2^{\text{eff}} \Delta_2 + \rho_2 A_2 W_2^{\text{eff}} \sigma_2^2 - \rho_2 B_2 W_2^{\text{eff}} (1-f)^2 \sigma_1^{22} = 0, \quad (18)$$

which yields the interior solution:

$$\alpha_2^* = \frac{\Delta_2 + \rho_2 \sigma_2^2 F_2}{\rho_2 W_2^{\text{eff}} \Sigma_2} \quad (19)$$

where $F_2 = W_2^{\text{eff}} + f \alpha_1 W_1^{\text{eff}}$ and $\Sigma_2 = \sigma_2^2 + (1-f)^2 \sigma_1^{22}$.

The solution is interior (i.e. $\alpha_2^* \in (0, 1)$) provided:

$$0 < \Delta_2 + \rho_2 \sigma_2^2 F_2 < \rho_2 W_2^{\text{eff}} \Sigma_2.$$

5.2.2 Optimal marketing m_2^*

Since player 1's allocation α_1 is already fixed, the fee income term $f \mu_2 \alpha_1 W_1^{\text{eff}}$ does not depend on m_2 . Marketing spend by player 2 only reduces W_2^{eff} , which is costly. Therefore:

$$m_2^* = 0. \quad (20)$$

5.3 Step 2 — Leader's Problem (Player 1)

Player 1 anticipates the follower's best response $\alpha_2^*(m_1)$ and solves:

$$\max_{\alpha_1, m_1} \mathcal{V}_1(\alpha_1, m_1; \alpha_2^*(m_1)).$$

5.3.1 Optimal allocation α_1^*

Symmetrically to equation (19):

$$\boxed{\alpha_1^* = \frac{\Delta_1 + \rho_1 \sigma_1^2 F_1}{\rho_1 W_1^{\text{eff}} \Sigma_1}} \quad (21)$$

where $F_1 = W_1^{\text{eff}} + f \alpha_2^* W_2^{\text{eff}}$ and $\Sigma_1 = \sigma_1^2 + (1-f)^2 \sigma_2^{12}$.

5.3.2 Derivative of follower's allocation with respect to m_1

The key object for the leader's marketing decision is how player 2's allocation responds to m_1 . This runs through the chain:

$$\frac{\partial \alpha_2^*}{\partial m_1} = \frac{\partial \alpha_2^*}{\partial \mu_1^2} \cdot \frac{\partial \mu_1^2}{\partial m_1} + \frac{\partial \alpha_2^*}{\partial \sigma_1^{22}} \cdot \frac{\partial \sigma_1^{22}}{\partial m_1}.$$

Define the marginal belief shift:

$$\Pi \equiv \frac{\partial \hat{p}_i^j}{\partial m_i} = \frac{\beta_i^{**}}{(\alpha_i^{**} + \beta_i^{**})^2} \cdot \frac{\gamma_i}{(m_i + \gamma_i)^2}. \quad (22)$$

Then:

$$\frac{\partial \mu_1^2}{\partial m_1} = (u_1 - d_1) \Pi, \quad (23)$$

$$\frac{\partial \sigma_1^{22}}{\partial m_1} = (u_1 - d_1)^2 (1 - 2\hat{p}_1^2) \Pi. \quad (24)$$

The partial derivatives of α_2^* with respect to μ_1^2 and σ_1^{22} are obtained by differentiating (19):

$$\frac{\partial \alpha_2^*}{\partial \mu_1^2} = \frac{(1-f)}{\rho_2 W_2^{\text{eff}} \Sigma_2} \equiv \frac{(1-f)}{D_2}, \quad (25)$$

$$\frac{\partial \alpha_2^*}{\partial \sigma_1^{22}} = -\frac{\alpha_2^* (1-f)^2}{\Sigma_2}, \quad (26)$$

where $D_2 \equiv \rho_2 W_2^{\text{eff}} \Sigma_2$.

Combining:

$$\boxed{\frac{\partial \alpha_2^*}{\partial m_1} = (u_1 - d_1) \Pi \left[\frac{(1-f)}{D_2} - \frac{\alpha_2^*(1-f)^2(u_1 - d_1)(1 - 2\hat{p}_1^2)}{\Sigma_2} \right]} \quad (27)$$

5.3.3 First-order condition for m_1^*

The leader's objective as a function of m_1 (with $\alpha_2^*(m_1)$ substituted in) is:

$$\begin{aligned} \mathcal{V}_1(m_1) = & \mu_1(W_1 - m_1) + \alpha_1(W_1 - m_1)[\mu_2^1(1-f) - \mu_1] \\ & + f\mu_1 \alpha_2^*(m_1) W_2 - \frac{\rho_1}{2} [A_1^2 \sigma_1^2 + B_1^2 \sigma_2^{12}]. \end{aligned} \quad (28)$$

The FOC $\partial \mathcal{V}_1 / \partial m_1 = 0$ gives:

$$f\mu_1 W_2 \cdot \frac{\partial \alpha_2^*}{\partial m_1} = \Psi, \quad (29)$$

where the right-hand side is the marginal cost of marketing:

$$\Psi \equiv \mu_1 + \alpha_1[\mu_2^1(1-f) - \mu_1] - \rho_1 A_1 W_1^{\text{eff}} \sigma_1^2 + \rho_1 B_1 W_1^{\text{eff}} (1-f)^2 \sigma_2^{12}. \quad (30)$$

Define:

$$\Phi \equiv f\mu_1 W_2 (u_1 - d_1) \left[\frac{(1-f)}{D_2} - \frac{\alpha_2^*(1-f)^2(u_1 - d_1)(1 - 2\hat{p}_1^2)}{\Sigma_2} \right]. \quad (31)$$

The FOC becomes $\Phi \cdot \Pi = \Psi$, i.e.:

$$\Phi \cdot \frac{\beta_1^{**}}{(\alpha_1^{**} + \beta_1^{**})^2} \cdot \frac{\gamma_1}{(m_1 + \gamma_1)^2} = \Psi.$$

Solving explicitly for m_1^* :

$$\boxed{m_1^* = \gamma_1 \left(\sqrt{\frac{\Phi \beta_1^{**} \gamma_1}{\Psi (\alpha_1^{**} + \beta_1^{**})^2}} - 1 \right)} \quad (32)$$

subject to $m_1^* \geq 0$, which requires $\Phi > \Psi$.

5.4 Equilibrium Summary

Proposition 5.1 (Stackelberg Equilibrium). *The equilibrium of the game is the quadruple $\{m_1^*, \alpha_1^*, m_2^* = 0, \alpha_2^*(m_1^*)\}$ where:*

$$m_2^* = 0, \quad (20)$$

$$m_1^* = \gamma_1 \left(\sqrt{\frac{\Phi \beta_1^{**} \gamma_1}{\Psi (\alpha_1^{**} + \beta_1^{**})^2}} - 1 \right), \quad (32)$$

$$\alpha_2^* = \frac{\Delta_2 + \rho_2 \sigma_2^2 F_2}{\rho_2 W_2^{\text{eff}} \Sigma_2}, \quad (19)$$

$$\alpha_1^* = \frac{\Delta_1 + \rho_1 \sigma_1^2 F_1}{\rho_1 W_1^{\text{eff}} \Sigma_1}. \quad (21)$$

6 Realised Portfolio Volatility

6.1 Equilibrium Variance Expression

Substituting equilibrium allocations into the variance of terminal wealth:

$$\mathcal{R}_i \equiv \text{Var}[W_i(T)] = \frac{\sigma_i^2 X_i^2 + \sigma_j^2 Y_i^2}{\Sigma_i^2}, \quad (33)$$

where σ_i^2 and σ_j^2 are *true* process variances (equations (2)), and:

$$X_i = F_i(1-f)^2 \sigma_j^{i2} - \frac{\Delta_i}{\rho_i} \quad (\text{own process loading}), \quad (34)$$

$$Y_i = \frac{(\Delta_i + \rho_i \sigma_i^2 F_i)(1-f)}{\rho_i} \quad (\text{cross process loading}). \quad (35)$$

6.2 Comparative Statics on Realised Volatility

The effect of each strategic variable on $\mathcal{R}_i$ is summarised below.

6.2.1 Effect of marketing m_i^*

Marketing affects $\mathcal{R}_i$ through two competing channels:

- (a) **Budget depletion:** $\partial F_i / \partial m_i = -1$, shrinking X_i and Y_i .
- (b) **Induced investment:** m_i raises α_j^* via the Jeffrey shift, increasing F_i through $f W_j^{\text{eff}} \partial \alpha_j^* / \partial m_i$.

The net effect is:

$$\frac{dF_i}{dm_i^*} = -1 + fW_j^{\text{eff}} \frac{\partial \alpha_j^*}{\partial m_i^*}.$$

Marketing *increases* $\mathcal{R}_i$ if and only if $fW_j^{\text{eff}} \partial \alpha_j^* / \partial m_i^* > 1$.

6.2.2 Effect of α_j^*

$\frac{\partial \mathcal{R}_i}{\partial \alpha_j^*} > 0$ always: more investment from player j amplifies player i 's exposure to their own process through fee income.

6.2.3 Effect of fee f

The fee enters through $B_i = \alpha_i W_i^{\text{eff}}(1 - f)$ and through Σ_i . Higher f reduces the cross-investment loading B_i but also reduces Σ_i , creating an ambiguous net effect on $\mathcal{R}_i$.

7 Structural GARCH Representation

7.1 Multi-Period Extension

Index periods by $t = 1, 2, \dots$. At each period the history updates: $k_i^t = k_i^{t-1} + \mathbf{1}_{r_i^t = u_i}$, $n_t = n_{t-1} + 1$. The Beta posterior updates accordingly:

$$\alpha_i^{*,t} = \alpha_i^0 + k_i^t, \quad \beta_i^{*,t} = \beta_i^0 + n_t - k_i^t.$$

7.2 Period t Surprise

Definition 7.1 (Return Surprise).

$$\epsilon_i^t = r_i^t - \mu_i = \begin{cases} (1 - p_i)(u_i - d_i) & \text{up move (prob. } p_i), \\ -p_i(u_i - d_i) & \text{down move (prob. } 1 - p_i). \end{cases}$$

Note $\mathbb{E}[\epsilon_i^t] = 0$ and $\mathbb{E}[(\epsilon_i^t)^2] = \sigma_i^2$.

7.3 Belief Dynamics

The posterior mean $\hat{p}_i^{j,t}$ updates as follows:

$$\Delta \hat{p}_i^{j,t} \equiv \hat{p}_i^{j,t} - \hat{p}_i^{j,t-1} = \frac{\mathbf{1}_{\text{up}} \beta_i^{**,t-1} - \mathbf{1}_{\text{down}} \alpha_i^{**,t-1}}{S_t(S_t + 1)}, \quad (36)$$

where $S_t = \alpha_i^{**,t-1} + \beta_i^{**,t-1}$ is the total pseudo-count and $\mathbf{1}_{up}, \mathbf{1}_{down}$ are indicators.

Taking the expectation-weighted approximation:

$$\Delta \hat{p}_i^{j,t} \approx \frac{\bar{C}_t \epsilon_i^t}{(u_i - d_i) S_t (S_t + 1)}, \quad (37)$$

where $\bar{C}_t = p_i \frac{\beta_i^{**,t-1}}{1 - p_i} + (1 - p_i) \frac{\alpha_i^{**,t-1}}{p_i}$ is an expected scaling factor.

7.4 Propagation to Realised Variance

Define the sensitivity of realised variance to the belief level:

$$\begin{aligned} \Lambda_t \equiv & (u_i - d_i) \frac{2(1-f)}{\rho_i \Sigma_{i,t}^2} [\sigma_j^2 Y_{i,t} - \sigma_i^2 X_{i,t}] \\ & + (u_i - d_i)^2 (1 - 2\hat{p}_t) \frac{2(1-f)^2}{\Sigma_{i,t}^3} [\sigma_i^2 X_{i,t} (F_{i,t} \Sigma_{i,t} - X_{i,t}) - \sigma_j^2 Y_{i,t}^2] \\ & + \frac{\partial \mathcal{R}_{i,t+1}}{\partial m_i^*} \cdot \frac{\partial m_i^*}{\partial \hat{p}_t}. \end{aligned} \quad (38)$$

Taking conditional expectations and using $\mathbb{E}[\epsilon_i^t | \mathcal{F}_t] = 0$, the linear term vanishes and the second-order expansion gives:

$$\mathbb{E}[\mathcal{R}_{i,t+1} | \mathcal{F}_t] = \mathcal{R}_{i,t} \left(1 - \frac{\kappa_t}{S_t} \right) + \theta_t (\epsilon_i^t)^2, \quad (39)$$

where:

$$\theta_t = \frac{\bar{C}_t^2}{2 S_t^2 (S_t + 1)^2} \frac{\partial \Lambda_t}{\partial \hat{p}_t} \quad (\text{ARCH coefficient}), \quad (40)$$

$$\kappa_t = \frac{\partial m_i^{*,t}}{\partial \hat{p}_i^{j,t}} \cdot \frac{\partial \hat{p}_i^{j,t}}{\partial \mathcal{R}_{i,t}} \quad (\text{mean-reversion coefficient}). \quad (41)$$

7.5 GARCH(1,1) Representation

Proposition 7.1 (Structural GARCH(1,1)). *Adding a baseline variance term $\omega_i^t = (\kappa_t/S_t) \bar{\mathcal{R}}_i$ where $\bar{\mathcal{R}}_i$ is the long-run variance, the conditional variance satisfies:*

$$\boxed{\mathbb{E}[\mathcal{R}_{i,t+1} | \mathcal{F}_t] = \underbrace{\omega_i^t}_{\text{baseline}} + \underbrace{\alpha_i^{\text{GARCH}} (\epsilon_i^t)^2}_{\text{ARCH term}} + \underbrace{\beta_i^{\text{GARCH}} \mathcal{R}_{i,t}}_{\text{GARCH term}}} \quad (42)$$

with:

$$\alpha_i^{\text{GARCH}} = \theta_t = \frac{\bar{C}_t^2(u_i - d_i)^2(1-f)}{\rho_i S_t^2(S_t + 1)^2 \Sigma_{i,t}^2} \left[\frac{\sigma_j^2 \rho_i \sigma_i^2 F_{i,t}}{\Sigma_{i,t}} + \frac{\sigma_j^2(1-f)}{\Sigma_{i,t}} - \frac{\sigma_i^2(1-f)^2 \sigma_j^2 \partial F_{i,t} / \partial \hat{p} - \sigma_i^2 \Delta_i / \rho_i}{\Sigma_{i,t}} \right] \quad (43)$$

$$\beta_i^{\text{GARCH}} = 1 - \frac{\kappa_t}{S_t} = 1 - \frac{\partial m_i^{*,t}}{\partial \hat{p}_i^{j,t}} \cdot \frac{\partial \hat{p}_i^{j,t}}{\partial \mathcal{R}_{i,t}}. \quad (44)$$

7.6 Endogenous GJR Asymmetry

The term $(1 - 2\hat{p}_t)$ in Λ_t (equation (38)) introduces a sign asymmetry between positive and negative surprises. This yields a GJR-GARCH type extension:

$$\mathbb{E}[\mathcal{R}_{i,t+1} \mid \mathcal{F}_t] = \omega_i^t + \alpha_i^{\text{GARCH}} (\epsilon_i^t)^2 + \gamma_i^{\text{GARCH}} (\epsilon_i^t)^2 \mathbf{1}_{\epsilon_i^t < 0} + \beta_i^{\text{GARCH}} \mathcal{R}_{i,t}, \quad (45)$$

where:

$$\gamma_i^{\text{GARCH}} = -\frac{\bar{C}_t^2(u_i - d_i)^2(1-f)^2(1-2\hat{p}_t)}{S_t^2(S_t + 1)^2 \Sigma_{i,t}^3} [\sigma_i^2 X_{i,t}(F_{i,t} \Sigma_{i,t} - X_{i,t}) - \sigma_j^2 Y_{i,t}^2]. \quad (46)$$

The sign of γ_i^{GARCH} depends on $\hat{p}_t \gtrless 1/2$:

- If $\hat{p}_t > 1/2$: negative surprises have *larger* volatility impact than positive — standard leverage effect.
- If $\hat{p}_t < 1/2$: positive surprises have larger impact — reversed asymmetry.

This asymmetry is *endogenous*: it emerges from the strategic marketing mechanism and Jeffrey updating rather than being imposed parametrically.

7.7 Stationarity Condition

Proposition 7.2 (Stationarity). *The GARCH process (42) is covariance-stationary if and only if:*

$$\alpha_i^{\text{GARCH}} + \beta_i^{\text{GARCH}} < 1 \iff \theta_t < \frac{\kappa_t}{S_t}.$$

This holds when the ARCH surprise effect is smaller than the mean-reversion force generated by the leader's marketing adjustment — a condition governed by the model's structural parameters.

8 Complete Parameter List

Table 1 lists every parameter and variable in the model.

Table 1: Complete parameter and variable list

Symbol	Description	Domain
<i>Primitives</i>		
W_i	Initial budget of player i	$\mathbb{R}_{>0}$
u_i	Up return of player i 's process	$\mathbb{R}$
d_i	Down return of player i 's process	$\mathbb{R}, d_i < u_i$
p_i	True up-probability (private to player i)	$(0, 1)$
f	Performance fee	$(0, 1)$
ρ_i	Risk aversion coefficient	$\mathbb{R}_{>0}$
α_i^0, β_i^0	Prior Beta pseudo-counts	$\mathbb{R}_{>0}$
γ_i	Marketing diminishing returns scale	$\mathbb{R}_{>0}$
<i>Observed history</i>		
n	Number of observed periods	$\mathbb{Z}_{>0}$
k_i	Number of up-moves in history	$\{0, \dots, n\}$
<i>Derived (Bayesian posterior)</i>		
$\alpha_i^* = \alpha_i^0 + k_i$	Posterior up pseudo-count	$\mathbb{R}_{>0}$
$\beta_i^* = \beta_i^0 + n - k_i$	Posterior down pseudo-count	$\mathbb{R}_{>0}$
<i>Decision variables</i>		
m_i	Marketing spend	$[0, W_i]$
α_i	Fraction allocated to other's process	$[0, 1]$
<i>Derived (Jeffrey shift)</i>		
$\delta(m_i) = m_i / (m_i + \gamma_i)$	Pseudo-count shift	$(0, 1)$
$\alpha_i^{**} = \alpha_i^* + \delta(m_i)$	Jeffrey-shifted up pseudo-count	$\mathbb{R}_{>0}$
$\beta_i^{**} = \beta_i^*$	Jeffrey-shifted down pseudo-count	$\mathbb{R}_{>0}$
$\hat{p}_i^j = \alpha_i^{**} / (\alpha_i^{**} + \beta_i^{**})$	Perceived up-probability	$(0, 1)$
$\mu_i^j = \hat{p}_i^j u_i + (1 - \hat{p}_i^j) d_i$	Perceived drift	$\mathbb{R}$
$\sigma_i^{j2} = (u_i - d_i)^2 \hat{p}_i^j (1 - \hat{p}_i^j)$	Perceived variance	$\mathbb{R}_{>0}$
<i>Equilibrium objects</i>		
$W_i^{\text{eff}} = W_i - m_i^*$	Effective budget	$\mathbb{R}_{>0}$
$\Delta_i = \mu_j^i (1 - f) - \mu_i$	Net return differential	$\mathbb{R}$
$\Sigma_i = \sigma_i^2 + (1 - f)^2 \sigma_j^{i2}$	Total perceived risk	$\mathbb{R}_{>0}$
$F_i = W_i^{\text{eff}} + f \alpha_j^* W_j^{\text{eff}}$	Fee-augmented budget	$\mathbb{R}_{>0}$
X_i, Y_i	Variance loadings (eqs. (34)–(35))	$\mathbb{R}$
$\mathcal{R}_i$	Realised portfolio variance	$\mathbb{R}_{>0}$
<i>Structural GARCH</i>		
$\epsilon_i^t = r_i^t - \mu_i$	Return surprise	$\mathbb{R}$
$S_t = \alpha_i^{**,t-1} + \beta_i^{**,t-1}$	Total pseudo-count	$\mathbb{R}_{>0}$
$\bar{C}_t$	Belief scaling factor	$\mathbb{R}_{>0}$
α_i^{GARCH}	ARCH coefficient	$\mathbb{R}_{>0}$

9 Guide to Symbolic Solution in SymPy / Mathematica

This section provides a complete ordered recipe for solving the model symbolically.

9.1 Step-by-Step Recipe

Step 1. Declare all primitives as symbols.

Parameters: $W_1, W_2, u_1, u_2, d_1, d_2, p_1, p_2, f, \rho_1, \rho_2, \alpha_1^0, \beta_1^0, \alpha_2^0, \beta_2^0, \gamma_1, \gamma_2, n, k_1, k_2$.

Step 2. Compute Bayesian posteriors.

$$\alpha_i^* \leftarrow \alpha_i^0 + k_i, \beta_i^* \leftarrow \beta_i^0 + n - k_i \text{ for } i = 1, 2.$$

Step 3. Declare m_1 as a symbol. Set $m_2 = 0$.

$$\text{Compute } \delta(m_1) = m_1/(m_1 + \gamma_1), \delta(m_2) = 0.$$

Step 4. Compute Jeffrey-shifted parameters.

$$\begin{aligned} \alpha_i^{**} &\leftarrow \alpha_i^* + \delta(m_i), \beta_i^{**} \leftarrow \beta_i^* \\ \hat{p}_i^j &\leftarrow \alpha_i^{**}/(\alpha_i^{**} + \beta_i^{**}). \end{aligned}$$

Step 5. Compute perceived drifts and variances.

$$\begin{aligned} \mu_i^j &\leftarrow \hat{p}_i^j u_i + (1 - \hat{p}_i^j) d_i \text{ (equation (5))}. \\ \sigma_i^{j2} &\leftarrow (u_i - d_i)^2 \hat{p}_i^j (1 - \hat{p}_i^j) \text{ (equation (6))}. \end{aligned}$$

Step 6. Compute true variances.

$$\sigma_i^2 \leftarrow p_i(1 - p_i)(u_i - d_i)^2 \text{ (equation (2))}.$$

Step 7. Declare α_1, α_2 as symbols and form F_i, Δ_i, Σ_i .

Equations (17), (15), (16). Note F_i depends on α_j .

Step 8. Solve the system $\{\alpha_1^* = \text{eq.}(21), \alpha_2^* = \text{eq.}(19)\}$ simultaneously.

This is a 2×2 linear system in (α_1, α_2) since F_1 depends on α_2 and F_2 depends on α_1 . Solve for $\alpha_1^*(m_1), \alpha_2^*(m_1)$ as functions of m_1 .

Step 9. Compute Π and $\partial\alpha_2^*/\partial m_1$.

Equations (22) and (27).

Step 10. Compute Φ and Ψ .

Equations (31) and (30).

Step 11. Solve the FOC $\Phi \cdot \Pi = \Psi$ for m_1^* .

Equation (32). Verify $m_1^* \geq 0$.

Step 12. Back-substitute m_1^* to get $\alpha_1^*, \alpha_2^*, W_i^{\text{eff}*}$.

Step 13. Compute realised variance $\mathcal{R}_i$.

Equations (34)–(33).

Step 14. Compute structural GARCH coefficients.

Declare ϵ_i^t, S_t as symbols.

Compute $\bar{C}_t, \Lambda_t, \theta_t, \kappa_t$.

Equations (43)–(46).

Step 15. Verify stationarity condition $\alpha_i^{\text{GARCH}} + \beta_i^{\text{GARCH}} < 1$.

9.2 Key Equation Cross-Reference

Table 2: Key equations for symbolic solution

Equation	Description	Number
μ_i, σ_i^2 from p_i, u_i, d_i	True process moments	(1), (2)
α_i^*, β_i^* from history	Bayesian posteriors	(3)
$\delta(m_i)$	Marketing shift	(22)
$\hat{p}_i^j, \mu_i^j, \sigma_i^{j2}$	Jeffrey-updated beliefs	(5)–(7)
α_2^*	Follower allocation	(19)
α_1^*	Leader allocation	(21)
m_1^*	Leader marketing	(32)
$\mathcal{R}_i$	Realised portfolio variance	(33)
GARCH(1,1)	Structural volatility dynamics	(42)
GJR extension	Asymmetric leverage	(45)

10 Comparative Statics Summary

Table 3: Comparative statics on perceived volatility σ_i^{j2}

Variable	Effect on σ_i^{j2}	Condition
$m_i \uparrow$	$\sigma_i^{j2} \uparrow$	if $\hat{p}_i^j < 1/2$
$m_i \uparrow$	$\sigma_i^{j2} \downarrow$	if $\hat{p}_i^j > 1/2$
$k_i \uparrow$	$\sigma_i^{j2} \uparrow$	if $\hat{p}_i^j < 1/2$
$k_i \uparrow$	$\sigma_i^{j2} \downarrow$	if $\hat{p}_i^j > 1/2$
$n \uparrow$	$\sigma_i^{j2} \downarrow$	if $\hat{p}_i^j < 1/2$
$n \uparrow$	$\sigma_i^{j2} \uparrow$	if $\hat{p}_i^j > 1/2$
$u_i - d_i \uparrow$	$\sigma_i^{j2} \uparrow$	unconditional

Table 4: Comparative statics on realised portfolio variance $\mathcal{R}_i$

Variable	Effect on $\mathcal{R}_i$	Condition
$m_i^* \uparrow$	$\mathcal{R}_i \uparrow$	if $fW_j^{\text{eff}} \partial \alpha_j^* / \partial m_i^* > 1$
$m_i^* \uparrow$	$\mathcal{R}_i \downarrow$	if $fW_j^{\text{eff}} \partial \alpha_j^* / \partial m_i^* < 1$
$\alpha_j^* \uparrow$	$\mathcal{R}_i \uparrow$	unconditional
$W_j^{\text{eff}} \uparrow$	$\mathcal{R}_i \uparrow$	unconditional
$\rho_i \uparrow$	$\mathcal{R}_i \downarrow$	unconditional
$f \uparrow$	ambiguous	depends on X_i, Y_i, Σ_i

Table 5: Structural GARCH parameter properties

Parameter	Economic source	Key driver
ω_i^t	Baseline process variance	True p_i , spread $u_i - d_i$
α_i^{GARCH}	Belief sensitivity to surprises	History length S_t , risk aversion ρ_i
β_i^{GARCH}	Marketing persistence	Budget constraints, $\partial m_i^* / \partial \hat{p}$
γ_i^{GARCH}	Asymmetric leverage	Position of $\hat{p}_t$ relative to $1/2$